

Assignment 3 : solns.

1. Find the energy levels of a particle moving in a potential field of the shape $V(x) = \begin{cases} \infty & \forall x < 0 \\ \frac{1}{2}\mu\omega^2 x^2 & \forall x \geq 0 \end{cases}$

Ans. The first step in solving this equation is to look at the boundary conditions. As $x \rightarrow 0$, the wave function should fall to zero. For $x > 0$, the wave function satisfies the differential equation for the harmonic oscillator. Since the odd wave functions for the harmonic oscillator tend toward zero as $x \rightarrow 0$, we can conclude that the equation for the odd states in the solution for the regular 1-D harmonic oscillator problem is the solution to the problem:

$$E = \hbar\omega \left(2n + \frac{3}{2}\right), \text{ where } n \text{ is a positive integer.}$$

2. Let u be a bound state eigenfunction of the Hamiltonian for a particle in a potential $V(\vec{r})$. Show that the expectation value vanishes for each component of linear momentum: $\langle u | p_i | u \rangle = 0$ Hint: consider $[x_i, \hat{H}]$.

Ans. Given: $\hat{H}u = Eu$

$$\langle u | [x_i, \hat{H}] | u \rangle = \langle u | x_i \hat{H} | u \rangle - \langle u | \hat{H} x_i | u \rangle = \langle u | x_i \hat{H} | u \rangle - \langle \hat{H}^\dagger u | x_i | u \rangle \stackrel{\hat{H}^\dagger = \hat{H}}{=} E (\langle u | x_i | u \rangle - \langle u | x_i | u \rangle) = 0$$

$$\text{But, } [x_i, \hat{H}] = \left[x_i, \left(\frac{\hat{p}^2}{2m} + V(\vec{r}) \right) \right] = \left[x_i, \frac{\hat{p}_i^2}{2m} \right] + [x_i, V(\vec{r})]$$

The first term has only p_i^2 , since x_i commutes with all $\hat{p}_j$ for $i \neq j$. Second term is zero.

$$\therefore [x_i, H] = \left[x_i, \frac{\hat{p}_i^2}{2m} \right] = \frac{1}{2m} ([x_i, \hat{p}_i] \hat{p}_i + \hat{p}_i [x_i, \hat{p}_i]) = i\hbar \frac{p_i}{m} \implies \langle u | p_i | u \rangle = \frac{m}{i\hbar} \langle u | [x_i, \hat{H}] | u \rangle = 0$$

3. Consider the case of a two-dimensional harmonic oscillator with the following Hamiltonian:

$$H_0 = \frac{p_x^2 + p_y^2}{2m} + \frac{k}{2}(x^2 + y^2)$$

Notice that the Hamiltonian is separable. Label the solutions as $|n_x, n_y\rangle$, where n_x and n_y are the quantum numbers corresponding to the oscillators in the x and y directions. What are the corresponding energies? What are the energy states (eigenfunctions) for this system?

Show that the states $\phi_{01} \equiv |0, 1\rangle$ and $\phi_{10} \equiv |1, 0\rangle$ are degenerate.

Ans. For separable Hamiltonian, the energy is the sum of energies for individual (separate) parts of the Hamiltonian

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The force constant for the oscillations in x and y are identical,

$$\therefore E_{m,n} = \left(m + \frac{1}{2}\right) \hbar\omega + \left(n + \frac{1}{2}\right) \hbar\omega = (m + n + 1)\hbar\omega, \text{ where } \omega = \sqrt{\frac{k}{m}}$$

Eigenstates (functions) : $u_{mn}(x, y) = N H_m(x) e^{-\frac{\xi_x^2}{2}} H_n(y) e^{-\frac{\xi_y^2}{2}}$; where $\xi_i = \frac{x_i}{x_0}$.

The energy for the given states are:

$$E_{01} = \left(0 + \frac{1}{2}\right) \hbar\omega + \left(1 + \frac{1}{2}\right) \hbar\omega = 2\hbar\omega = E_{10}$$

i.e., $\phi_{01} \equiv |0, 1\rangle$ and $\phi_{10} \equiv |1, 0\rangle$ are degenerate.

4. The Cartesian components $\hat{L}_i$; $i = 1, 2, 3$ of the ang. momentum operator are related to the Cartesian components of the position and momentum operators by $\hat{L}_i = \epsilon_{ijk} x_j p_k$. (a) Does the order of the operators on the right matter? (b) Show:

$$[\hat{L}_i, x_j] = i\hbar\epsilon_{ijk}x_k; \quad [\hat{L}_i, p_j] = i\hbar\epsilon_{ijk}p_k; \quad [\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k$$

Ans. $\therefore$ in $\hat{L}_i = \epsilon_{ijk}x_j p_k$; The indices j and k have to be different on the right, and then x_j and p_k commute, so the order of the operators does not matter.

(b) $[L_i, x_j] = \epsilon_{ikl} [x_k p_l, x_j] = \epsilon_{ikl} \{x_k [p_l, x_j] + [x_k, x_j] p_l\}$ The second term is zero and the first term is $-i\hbar\delta_{lj}$.
 $= -i\hbar\epsilon_{ikj}x_k = i\hbar\epsilon_{ijk}x_k$ similarly for $[L_i, p_j]$

For $[L_i, L_j]$ one can use the compact way, but it is easier to do it longhand.

$$[L_1, L_2] = [x_2 p_3 - x_3 p_2, x_3 p_1 - x_1 p_3] = [x_2 p_3, x_3 p_1] + [x_3 p_2, x_1 p_3] + 0 + 0$$

$$= i\hbar(x_1 p_2 - x_2 p_1) + 0 + 0 = i\hbar L_3$$

$$[L_i, L_i] = 0 \forall i \text{ and } [L_2, L_1] = -[L_1, L_2] = -i\hbar L_3$$

Keeping in mind that the directions x_i , $i = 1, 2, 3$ are arbitrary and as long as the relative orientations are in the same cyclic order, we can exchange $i \leftrightarrow j \leftrightarrow k \leftrightarrow i$. $\therefore [\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k$.

5. For a 1-dim harmonic oscillator, the spring constant changes from k to $k(1 + \epsilon)$, where ϵ is small. Calculate the first order perturbation in the energy.

Ans. We can calculate it trivially, by writing

$$E_v = \frac{\hbar\omega_{\text{new}}}{2} \left(v + \frac{1}{2}\right) = \frac{\hbar}{2} \left(v + \frac{1}{2}\right) \sqrt{\frac{k(1+\epsilon)}{\mu}} = \frac{\hbar\omega}{2} \left(v + \frac{1}{2}\right) [1 + \frac{\epsilon}{2} + \mathcal{O}(\epsilon^2)]$$

and $\therefore$ first order perturbation in the energy $= \epsilon \frac{\hbar\omega}{2} \left(v + \frac{1}{2}\right)$

It is interesting to see that if we use the bound state perturbation theory, we get the same result.

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\mu\omega^2 x^2 = \hbar\omega \hat{a}^\dagger \hat{a} + \frac{\hbar\omega}{2}, \text{ where } \hat{a} = \frac{1}{\sqrt{2x_0}} \left(\hat{x} + \frac{i}{\mu\omega} \hat{p}\right); x_0 = \sqrt{\frac{\hbar}{\mu\omega}} \text{ and } \hat{a}^\dagger \text{ is its adjoint.}$$

$$\omega = \sqrt{\frac{k}{\mu}}. \text{ Thus for } k \text{ changed to } k(1 + \epsilon), \omega_{\text{new}} = \sqrt{\frac{k(1+\epsilon)}{\mu}},$$

$$\therefore \hat{H}' = \frac{1}{2}\mu\omega_{\text{new}}^2 x^2 - \frac{1}{2}\mu\omega^2 x^2 = \frac{1}{2}\epsilon\mu\omega^2 x^2$$

The zeroth order energy levels are: $E_v = \left(v + \frac{1}{2}\right)\hbar\omega$

The first order corrections are:

$$H'_{v,v} = \langle v | H' | v \rangle = \frac{1}{2}\epsilon\mu\omega^2 \langle v | x^2 | v \rangle = \frac{1}{2}\epsilon\mu\omega^2 \cdot \frac{x_0^2}{2} \langle v | (\hat{a} + \hat{a}^\dagger)^2 | v \rangle$$

using $\hat{a}|v\rangle = \sqrt{v}|v-1\rangle$ and $\hat{a}^\dagger|v\rangle = \sqrt{v+1}|v+1\rangle$, for orthonormal basis of solution states $|v\rangle$, we have:

$$\begin{aligned}
H'_{v,v} &= \frac{\epsilon \hbar \omega}{4} [\langle v | (\hat{a} + \hat{a}^\dagger) (\sqrt{v} |v-1\rangle + \sqrt{v+1} |v+1\rangle)] = \\
&= \frac{\epsilon \hbar \omega}{4} \left[\langle v | \left\{ \sqrt{v(v-1)} |v-2\rangle + v |v\rangle + (v+1) |v\rangle + \sqrt{(v+1)(v+2)} |v+2\rangle \right\} \right] = \frac{\epsilon \hbar \omega}{4} (2v+1) = \epsilon \frac{\hbar \omega}{2} (v + \frac{1}{2})
\end{aligned}$$